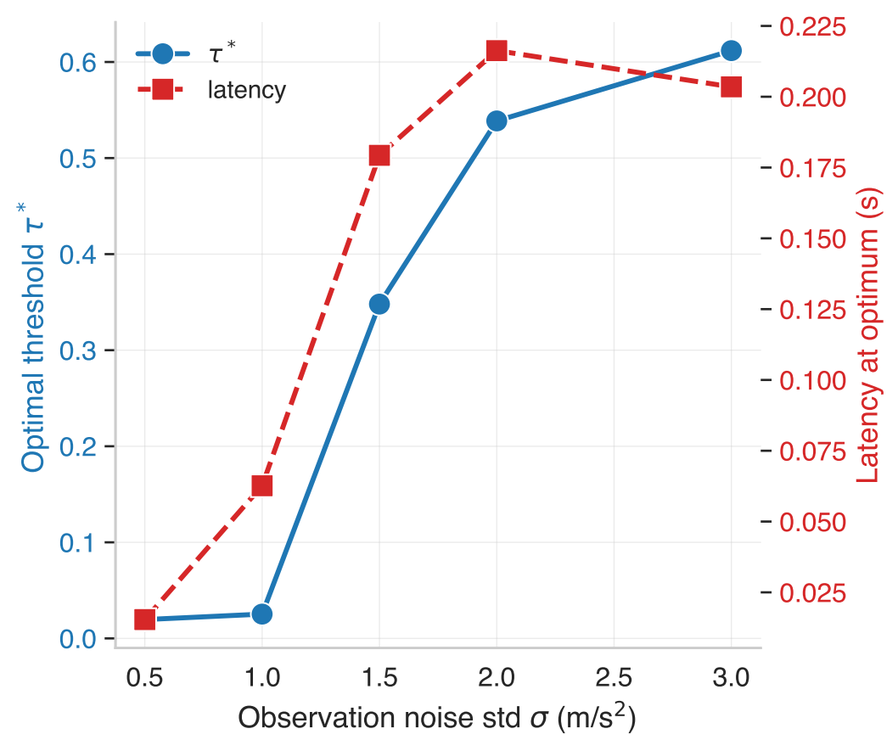
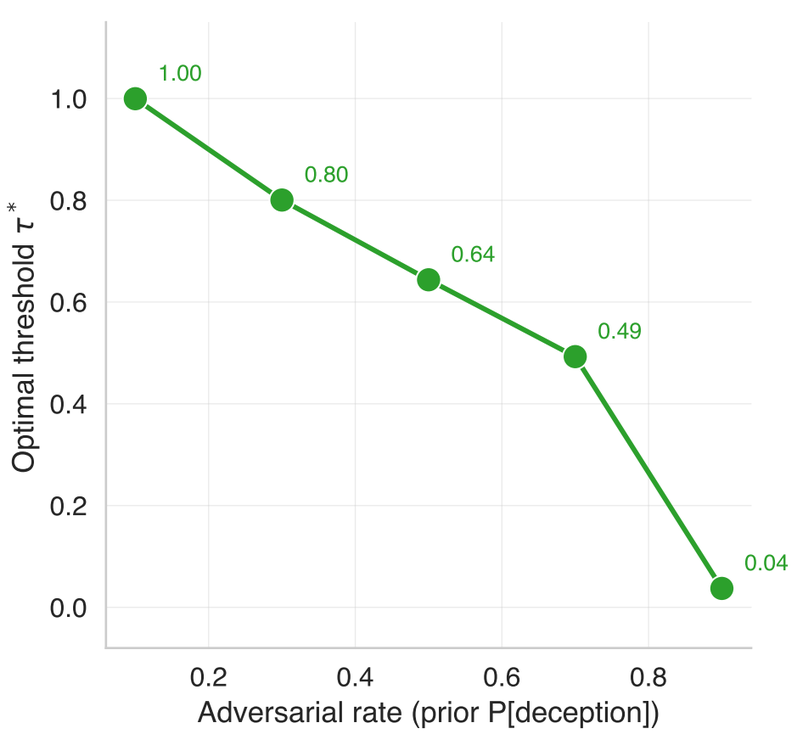
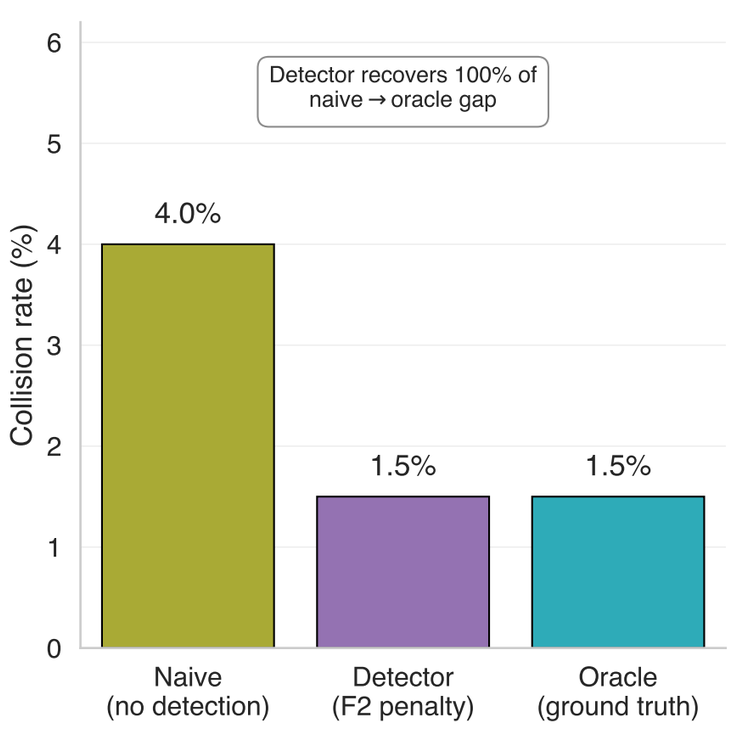

τ*
latency
Optimal threshold τ*
Latency at optimum (s)
Observation noise std σ (m/s²)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.025
0.050
0.075
0.100
0.125
0.150
0.175
0.200
0.225
0.5
1.0
1.5
2.0
2.5
3.0


Optimal threshold τ*
Adversarial rate (prior P[deception])
1.00
0.80
0.64
0.49
0.04
0.0
0.2
0.4
0.6
0.8
1.0
0.2
0.4
0.6
0.8


Detector recovers 100% of naive → oracle gap
Collision rate (%)
4.0%
1.5%
1.5%
Naive (no detection)
Detector (F2 penalty)
Oracle (ground truth)
0
1
2
3
4
5
6